

NBA Free Throw Prediction Using Machine Learning Classification Models

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Abstract—Sports analytics has become one of the most important applications of machine learning in modern sports. Basketball organizations increasingly rely on data-driven systems to evaluate player performance, optimize strategies, and improve decision-making.

This project presents a machine learning-based NBA free throw prediction system capable of classifying whether a player performs above or below the league-average free throw percentage of their NBA season using season-long basketball statistics.

The dataset was collected through web scraping from Basketball Reference and contains NBA player statistics from seasons between 1975 and 2026.

Several machine learning classification algorithms were trained and evaluated, including Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Classifier (SVC), Random Forest, and Gradient Boosting.

Experimental results showed that Logistic Regression achieved the highest accuracy with 90.72%.

The project was extended in Phase 2 through the development of an interactive Streamlit web application that allows users to input player statistics and receive real-time predictions.

Index Terms—Machine Learning, Sports Analytics, NBA, Classification, Free Throw Prediction

I. INTRODUCTION

Machine learning has become one of the most influential technologies in sports analytics. Basketball organizations use machine learning techniques to evaluate players, improve team strategies, and predict player performance.

Free throw shooting efficiency is one of the most critical aspects of basketball performance. Free throws can determine the outcome of close games, especially during clutch moments.

This project aims to build a machine learning classification system capable of predicting whether an NBA player performs above or below the league-average free throw percentage for their season based on season-long basketball statistics.

The project was divided into two phases:

- Phase 1: Data collection, preprocessing, feature engineering, and model training.
- Phase 2: Model evaluation, visualization, and deployment using Streamlit.

II. PROBLEM STATEMENT

The objective of this project is to classify NBA players into two categories:

- Players performing above the league-average free throw percentage for their season
- Players performing below the league-average free throw percentage for their season

Machine learning allows deeper analysis by identifying hidden relationships between player statistics and free throw performance.

The system aims to support:

- Basketball coaches
 - Analysts
 - Scouts
 - Sports organizations
- through data-driven decision making.

III. DATASET DESCRIPTION

The dataset was collected using web scraping from Basketball Reference.

A. Data Source

- <https://www.basketball-reference.com>
- <https://www.basketball-reference.com/leagues/>

B. Dataset Overview

TABLE I
DATASET OVERVIEW

Property	Value
Dataset Source	Basketball Reference
Timeframe	1975 – 2026
Rows	26,292
Columns	32
Task Type	Binary Classification

The dataset contains player season averages and basketball statistics such as:

- Games Played (G)
- Games Started (GS)
- Minutes Played (MP)
- Field Goal Percentage (FG%)
- Three Point Percentage (3P%)
- Two Point Percentage (2P%)
- Effective Field Goal Percentage (eFG%)
- Rebounds (TRB)

- Assists (AST)
- Steals (STL)
- Blocks (BLK)
- Turnovers (TOV)
- Personal Fouls (PF)
- Points Per Game (PTS)

IV. PROJECT WORKFLOW

The workflow of the project consisted of several stages:

- 1) NBA data web scraping
- 2) Data preprocessing
- 3) Feature engineering
- 4) Feature scaling
- 5) Model training
- 6) Model evaluation
- 7) Streamlit deployment

V. DATA PREPROCESSING

A. Handling Missing Values

Rows containing missing values were removed.

The Awards column contained more than 23,000 missing values and was dropped entirely.

B. Removing Unnecessary Features

The following columns were removed:

- RK
- Player Name

C. Duplicate Player-Season Handling

Some players appeared multiple times in the dataset due to mid-season transfers between teams.

To ensure consistency, duplicate player-season entries were removed by keeping one representative row for each player and season combination.

D. Feature Engineering

Initially, the project classified players using a fixed threshold where:

- $FT\% \geq 0.75 \rightarrow$ Above Average
- $FT\% < 0.75 \rightarrow$ Below Average

However, this methodology was later improved because NBA league-average free throw percentages vary significantly across different eras.

Instead of using a fixed threshold, the final methodology compared each player's FT% against the league-average FT% of the same season.

League-average FT% values were extracted and merged into the dataset during preprocessing.

Rows with missing FT% values were removed because they could not form valid targets.

The final target variable was generated using:

$$Target = \begin{cases} 1, & \text{if } FT\% \geq \text{League Average } FT\% \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

This methodology created a more adaptive and historically fair classification system because players were evaluated relative to the shooting standards of their own NBA season rather than using a universal threshold.

After target generation, the class distribution became approximately balanced:

- Class 0: 53.7%
- Class 1: 46.3%

This improved model stability and reduced classification bias during training.

E. Train-Test Split

The dataset was divided into:

- 80% training data
- 20% testing data

F. Feature Scaling

Feature scaling was applied using StandardScaler for models sensitive to feature magnitude such as KNN and SVC.

The scaling equation is:

$$Z = \frac{X - \mu}{\sigma} \quad (2)$$

Where:

- X = original value
- μ = mean
- σ = standard deviation

VI. MACHINE LEARNING MODELS

Five machine learning classification algorithms were implemented and evaluated.

A. Logistic Regression

Logistic Regression achieved the highest performance among all models with 90.72% accuracy.

Hyperparameters used:

- Solver = liblinear
- Maximum iterations = 1000

B. K-Nearest Neighbors

KNN predicts performance based on similarity between neighboring players.

Hyperparameters used:

- Number of neighbors = 5

C. Support Vector Classifier

SVC creates strong decision boundaries between classes.

Hyperparameters used:

- Kernel = RBF
- $C = 1.0$

D. Random Forest

Random Forest captures nonlinear relationships using decision trees.

Hyperparameters used:

- Number of estimators = 100
- Random state = 42

E. Gradient Boosting

Gradient Boosting improves performance iteratively by learning from prediction errors.

Hyperparameters used:

- Number of estimators = 100
- Learning rate = 0.1

VII. MODEL EVALUATION

Accuracy was used as the primary evaluation metric.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

Where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

Additional evaluation metrics included:

- Precision
- Recall
- F1-Score

VIII. EXPERIMENTAL RESULTS

TABLE II
MODEL ACCURACY COMPARISON

Model	Accuracy (%)
Logistic Regression	90.72
KNN	71.48
SVC	82.75
Random Forest	77.16
Gradient Boosting	85.17

IX. CLASSIFICATION METRICS

TABLE III
CLASSIFICATION METRICS

Model	Precision	Recall	F1-Score
Logistic Regression	0.91	0.91	0.91
KNN	0.72	0.71	0.71
SVC	0.83	0.83	0.83
Random Forest	0.77	0.77	0.77
Gradient Boosting	0.85	0.85	0.85

X. CONFUSION MATRICES

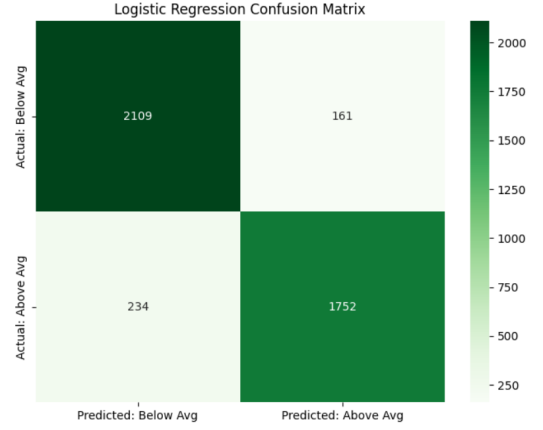


Fig. 1. Logistic Regression Confusion Matrix

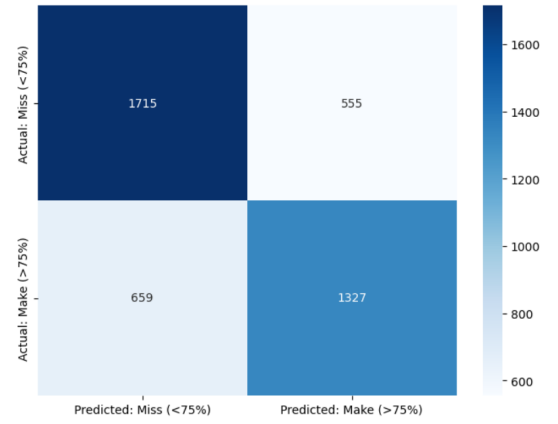


Fig. 2. KNN Confusion Matrix

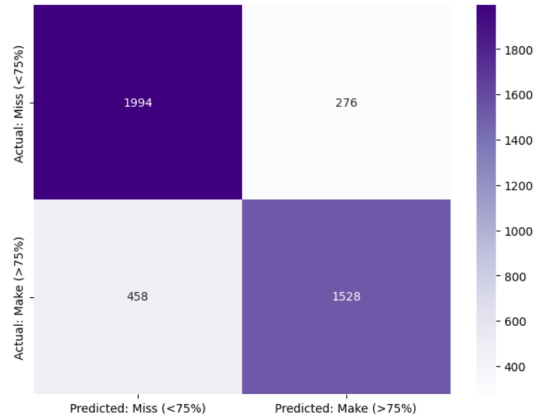


Fig. 3. SVC Confusion Matrix

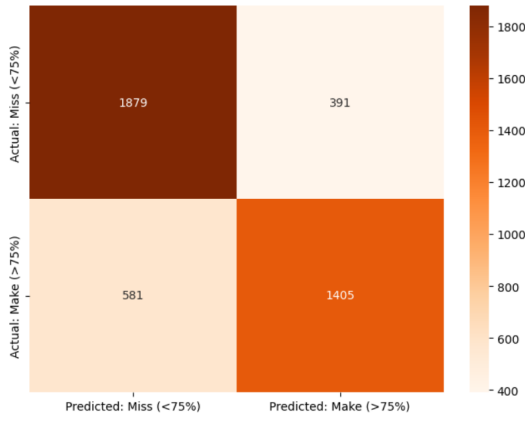


Fig. 4. Random Forest Confusion Matrix

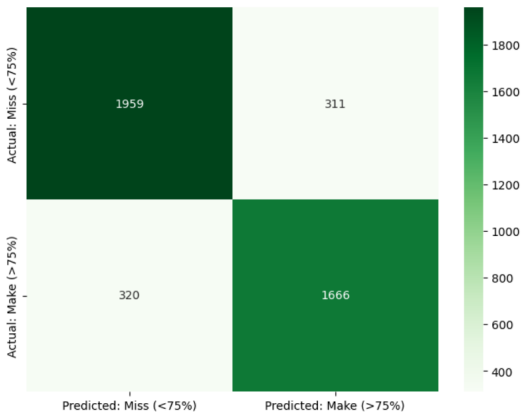


Fig. 5. Gradient Boosting Confusion Matrix

XI. PHASE 2: STREAMLIT DEPLOYMENT

In Phase 2, an interactive Streamlit web application was developed.

The application allows users to:

- Select a machine learning model
- Input player statistics
- Generate predictions
- Compare model performance

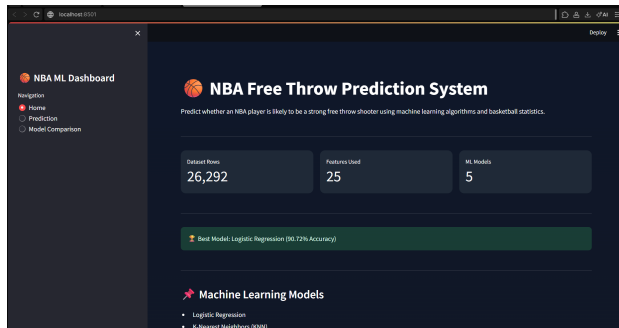


Fig. 6. Streamlit Home Page

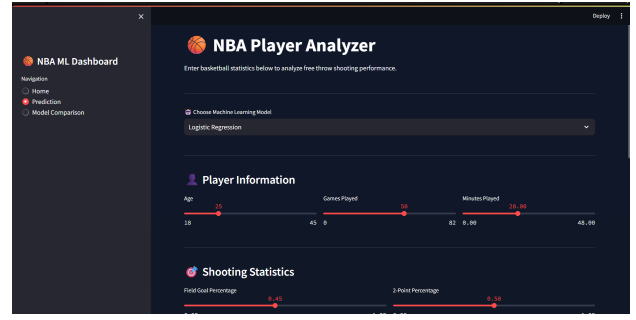


Fig. 7. Prediction Dashboard

XII. DISCUSSION

The project demonstrated that machine learning can effectively predict NBA free throw performance using player statistics.

Logistic Regression achieved the highest accuracy among all tested models. This indicates that the relationship between basketball shooting statistics and free throw efficiency is relatively structured and linearly separable.

Feature scaling significantly improved KNN and SVC performance because these algorithms depend heavily on geometric distance calculations.

KNN produced the lowest performance because high-dimensional basketball statistics can reduce neighbor quality and increase classification difficulty.

Tree-based models such as Random Forest and Gradient Boosting successfully captured nonlinear relationships between basketball statistics. However, they did not outperform Logistic Regression because the dataset contained highly correlated statistical features that favored simpler linear decision boundaries.

The results also demonstrated strong relationships between:

- Three-point shooting and free throw shooting
- Effective field goal percentage and offensive efficiency
- Offensive consistency and shooting discipline

XIII. LIMITATIONS

Several limitations were identified during the project:

- Per-game averages may hide consistency variations
- Psychological pressure factors were not included
- Injury history was not considered
- Training conditions and coaching styles were unavailable
- NBA playstyles changed significantly between 1975 and 2026
- Rule changes across eras may affect statistical consistency

XIV. CONCLUSION

This project presented a machine learning-based NBA free throw prediction system using historical basketball statistics.

The dataset underwent preprocessing, feature engineering, feature scaling, and model training using multiple machine learning algorithms.

Experimental results showed that Logistic Regression achieved the highest performance with 90.72% accuracy.

The project was successfully deployed using Streamlit to provide real-time predictions and model comparison.

Overall, the project demonstrates the effectiveness of machine learning in sports analytics applications.

XV. FUTURE WORK

Future improvements may include:

- Using advanced NBA statistics
- Hyperparameter optimization
- Deep learning implementation
- Real-time NBA API integration
- Cloud deployment
- Advanced visualization dashboards

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